Nonconformity management process

ram-sCHIP
FNALOGIXEDIGNAL

Document no	**RS-SP-11**	**Written by**	Mark.Kim
Rev no	**001**		
Rev. date	2022.09.26	**Department**	QM

Revised contents	**Drift**	**Review**	**Approval**
	/	/	/

Revised Items	**AS IS**	**TO BE**

Agreement	**QM**	**R&D**					

Confidential

	Opinion:

ram-sCHIP
FNALOGIXEDIGNAL

[Revision history]

Revision history summary			RS-SP-11	
			Revision	001
			Confidential	G1
Rev. no	**Rev. date**	**Simple description of changed contents**	**Author**	**Post**
001	2022.09.26	Newly established.	Mark.Kim	QM

Confidential

1. Purpose and scope

This process is issued to prevent shipment to the customers in advance in case of nonconformity that does not satisfy the required quality in the production process, and to define root cause of nonconformity and countermeasure for root cause.

Applied scope : all of process (subcontractor's process, ramschip's design)

2. Definition of terms

2.1 Nonconformity

The products which are not meet to the customer requirement and product own specification.

2.2 Low yield

It means a decrease in yield caused by quality issue or any problem occurring during the process.

Generally, a low yield limit is defined by average yield for 6months.

2.3 Corrective action

Applying countermeasures against the cause of nonconformity to the manufacturing process.

2.4 Rework

The actions to ensure that nonconformity products is able to be satisfied the required spec.

2.4 Return

Sending abnormal product to manufacturing process.

2.5 Scrap

Disposal because it can't be reworked.

2.6 Concession

The using of the nonconformity products without repair or rework if nonconformity is not included critical characteristic of the product. or it is acceptable by the customer's approval

3. Relevant form

3.1 Abnormal Lot report(ANR) [RS-SP-11-F001]

3.2 Nonconformity Notification report(NCR) [RS-SP-11-F002]

3.3 Nonconformity history & LOT List [RS-SP11-F003]

Confidential

4. Responsibility and authority

4.1 Subcontractor

Subcontractor should report ramschip if it's occurred to nonconformity issue during manufacturing Immediately. and, should stop mass production until be done root cause definer and corrective action. The root cause & corrective actions have to reasonable & clear. And subcontractor have to submit a verification and effectiveness report to ramschip.

4.2 Ramschip QM div

The CQO have responsibility for handling of the all of nonconformity issues.
If it's happened nonconformity issue from subcontractors or customers, CQO have to identify of the affected range immediately. And then, should request corrective actions and countermeasure to the subcontractors. And, CQO have authority regarding disposition of nonconformity product.

4.3 PE Team (Product engineering)

If the Nonconformity is issued from subcontractor to ramschip's QM, PE team should find actual root cause for nonconformity issue with subcontractors. And then, should report QM analysis result.

5. Process mapping

Confidential

Work flow	Who
Nonconformity issue	Subcontractors
↓	
Reporting to ramschip (To QM)	Subcontractors
↓	
Survey affect range	Subcontractors / QM
↓	
Root casue define/ Corrective action ← NG	Subcontractors / PE
↓	
Feasibility review for action	QM / PE
↓ OK	
Affect lot disposition	QM
↓	
Monitoring of affect Lot	Subcontractors / QM
↓	
Customer issue monitoring after shipping	QM

Com

Phase	Input	Process	Output	Process owner	Support team
P1	Nonconformity issue	-. Occurrence of the nonconformity during manufacturing. -. Stop to mass production. -. Reporting of the nonconformity issue & status to the ramschip's QM. -. In case of TEST low yield, it does not stop mass production. -. Should be surveyed to the affect range clearly. -. The nonconformity product must be isolated from the normal product.	-. Abnormal Lot report (ANR) -. Nonconformity notification report (NCR)	Subcontractors.	QM
P2	-. ANR -. NCR	-. QM investigate quality risk for the issued nonconformity. -. If there is critical quality risk, request reliability test or evaluation for nonconformity product to the issued site.	Risk investigation	QM	Subcon
P3	-. ANR -. NCR	-. Subcontractor submit ramschip update report which is defined root cause and corrective actions.	-. Update ANR/NCR	Subcon	QM PE
P4	-. Update ANR/NCR	-. QM division review update report from subcontractors. And then, QM division Review update actions whether it is reasonable or not. -. If it is reasonable, QM charged person judge LOT disposition. -. But, in case of not reasonable them, should request again for Root cause define & corrective action to subcontractors. -. If it needs to customer notification for nonconforming Issue, QM engineer should report them to the customer.	-. LOT disposition -. Customer Notification	QM	Subcon Customer
P5	Lot disposition	-. Subcontractors should handled according to the instruction of the ramschip's QM engineer.	-. Lot handling	QM	Subcon
P6	Lot handling	-. QM engineer should perform to the tracking from issued process to shipping for the nonconforming product. -. And, should monitor quality issue from field after shipping. -. QM engineer should record for nonconforming history and issued Lot list.	-. Nonconformity lot list.	QM	Subcon

6. Record

Material	Form code	Dept	Life

Confidential

Abnormal Lot report (ANR)	RS-SP-11-F001	QM	5 Years.
Nonconformity notification report(NCR)	RS-SP-11-F002	QM	5 Years.
Nonconformity history & LOT List	RS-SP-11-F003	QM	5 Years.

Confidential

Appendix

Confidential

LOW Yield LOT REPORT (ANR)

ram-sCHIP
FNALOGIXEDIGNAL

I. General information [TEST Site]

Company		Date	
Engineer			

II. LOW Yield information [TEST Site]

Device		Process (PT/ FT_hot/ FT_Room)	
Wafer run ID		LOT ID	
Input Q'ty		Output Q'ty / (Yield %)	/
SBL Limit (%)		Yield of issued BIN (%)	
Main fail category	Main fail category		
Data-log file		Shmoo	

III. Fail analysis [Ramschip]

Reason of low yield	
Countermeasure	

IV. LOT Disposition [ramschip QM]

Lot disposition		Charge person	
Special comment (To subcontractors)			

V. Abnormal LOT handling

LOT Handling		Engineer	

Confidential

Engineer comment	

Confidential

Nonconformity notification report(NCR)

ram-sCHIP
FNALOGIXEDIGNAL

I. General information [subcontractors]

Company		Date	
Engineer			

II. Nonconformity information [subcontractors]

Device		Wafer run ID	
Wafer slot information (If necessary)		LOT ID	
Issue process		Yield %(In/Out Q'ty)	
Nonconformity issue			
Nonconforming mode	☐ Critical defect issue (Special characteristics) _ Chip/ Package crack, Bond pull, Ball shear, Ball crack, Wire stitch crack, Dimension issue, Wire short, Pad crack after bond pull test, chipping (attack to active pattern) ☐ Yield attack caused by visual defects or loss. (Accumulated yield 3%down each process)		
Root cause			
Corrective action			

III. LOT Disposition [ramschip's QM]

Lot disposition notice		Charge person	
QM Comment			

IV. LOT Handling [Subcontractors]

LOT Handling		Engineer	

Confidential

Engineer comment	

Confidential

Nonconformity history & LOT List

ram-sCHIP
ANALOG MIXED SIGNAL

Device	Wafer ID	LOT ID	Site/Process	Nonconforming issue	Affect LOT info.	Disposition	Issue date

Confidential

Confidential